

Prime-Ablate-Annihilate Protocol

FINALIZED BIOPHYSICAL FRAMEWORK FOR EARLY-STAGE OSCC (V7)

Executive Summary

This protocol provides a definitive, closed-loop biophysical intervention for Early-Stage Oral Squamous Cell Carcinoma (OSCC). Replacing broad mechanical resection with **in-situ sterilization**, it leverages differential homeostatic resilience between malignant and healthy tissues. By utilizing Nernstian ionic priming, Gold Nanostar (GNS)-enhanced photothermal therapy, and real-time impedance-gated nsPEF/CAP ablation, the protocol ensures complete tumor annihilation while preserving vital mucosal architecture and function.

1. Patient Selection & Case Criteria

Targeting the "ideal" 15-25% of presenting cases to maximize functional preservation and oncological safety.

- **Primary Staging:** T1–T2, N0, M0.
- **Depth of Invasion (DOI):** ≤ 4 mm.
- **Exclusion:** Cardiac devices (nsPEF interference); T3–T4/N+ disease.
- **Subsite Specialization (Floor of Mouth):** For DOI ≥ 4 mm, Elective Selective Neck Dissection (SND I-III) is mandatory due to SLNB shine-through.

2. Step-by-Step Clinical Workflow

Stage 0A: Hyperoxic Prep

Hyperbaric Oxygen Therapy (2.0 ATA, 60 min) to downregulate HIF-1 α . Post-egress 5-minute hyperoxic window necessitates active phase initiation within 7 minutes.

Stage 0B: Priming & Thermal Sensitization

Application of Chitosan-Poloxamer hydrogel loaded with **Gold Nanostars (GNS)**. Irradiation (810 nm) induces localized volumetric heating ($\geq 43^{\circ}\text{C}$ at 4 mm) via GNS-enhanced photothermal conversion, alongside Nernstian priming (Low- K^{+} 1 mM / High- Ca^{2+} 2 mM) for membrane stabilization.

Stage 1: Closed-Loop Ablation

Automated nsPEF delivery utilizing a diagnostic pre-sweep to scale pulse amplitude (10–60 kV/cm) to tissue impedance. This guarantees dielectric breakdown regardless of local tumor heterogeneity.

Stage 2: Annihilation

Cold Atmospheric Plasma (CAP) Jet application for RONS-mediated surface and subsurface annihilation, **gated by impedance saturation ($d\sigma/dt = 0$)**.

3. Operational Parameter Matrix

Protocol Phase	Target/Constraint	Mechanism
0A: Prep	2.0 ATA / 60 min	HIF-1 α downregulation.
0B: Prime	810 nm + GNS Gel	GNS-PTT; Nernstian fail-safe.
1: Ablation	Closed-Loop nsPEF	Impedance-gated scaling.
2: Annihilation	CAP Jet + RONS	$d\sigma/dt = 0$ Gate.

4. Clinical Workflow Strategy

Tier	Decision Metric	Strategy
Primary Ablation	T1/T2, DOI $\leq$ 4 mm	Ex-vivo biopsy; serial surveillance.
Ablative Adjuvant	T2/T3, DOI $\geq$ 4 mm	Histopathological R-classification.